AUSTRIA-QD: A Quarterly Database for Macroeconomic Research in Austria

Franz X. Mohr

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Abstract

This report's contribution is a *methodology*, not a dataset: a documented, live-verified, and fully reproducible procedure for building a FRED-QD-style (McCracken and Ng, 2020) quarterly macroeconomic panel for a single country from public international statistical APIs, and its reference implementation for Austria. Larger academic releases already cover Austria as part of a multi-country panel – notably EA-MD-QD (Barigozzi and Lissona, 2024), with 1,136 series for the euro area and its ten largest members – so this report does not compete on coverage. Instead it demonstrates, and documents in enough detail to be repeated for any other country, a construction discipline that a periodically-released static file cannot easily expose: every source (Eurostat, the OECD, the IMF, the ECB, the BIS, and, for two country-specific overrides, the European Commission's Business and Consumer Survey and Yahoo Finance) is confirmed against its live API – structure query, then a real data pull returning current, plausible values – rather than assumed correct from documentation, with the exact identifier and any conceptual caveat recorded in a public, machine-readable registry rather than left implicit in code or spreadsheet formulas. One concrete payoff of this discipline: merging rather than strictly falling back between sources for the anchor national-accounts concepts recovers 35 years of Austrian history that a naive Eurostat-then-OECD fallback would silently forfeit, extending coverage to 1960-Q1 – four decades earlier than EA-MD-QD's stated January 2000 start. A second payoff came from a country-agnostic plausibility-check layer added specifically so a researcher extending this project to a new country has some automated verification beyond eyeballing a spreadsheet: on its first live run, it caught a previously-unknown ∼4x level discontinuity that very merge had introduced (OECD's contribution was on an annualized-rate scale, Eurostat's was not), invisible to growth-rate correlation checks alone – fixed the same day it was found. The reference implementation itself covers 38 concepts, identical in column structure for every country the underlying open-source tool has been run for, and the report catalogs, series by series, which of FRED-QD's remaining 207 concepts have a plausible but as yet unverified Austrian counterpart, which are directly computable from concepts already implemented, and which have no cross-country equivalent at all – itself a template for scoping the same exercise elsewhere.

JEL Classification: C82, C33, E01

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1 Introduction

McCracken and Ng (2020) introduced FRED-QD as the quarterly companion to FRED-MD (McCracken and Ng, 2016), itself designed to emulate the macroeconomic panel used by Stock and Watson (2012). Their stated goal was narrow but consequential: relieve researchers working with macroeconomic "big data" methods of the burden of assembling, revising, and documenting a large panel themselves, by providing one standardized, continuously updated, publicly available file. FRED-QD has since become a standard input to factor models, large Bayesian VARs, and a range of other data-rich empirical methods – and has itself inspired country- and region-specific analogs, most notably a Canadian version (Fortin-Gagnon et al., 2018) and, for the euro area, Barigozzi and Lissona (2024)'s EA-MD-QD, which provides 1,136 monthly and quarterly series for the euro area as a whole and its ten largest member states, Austria included, drawn from Eurostat, the ECB, the OECD, and FRED.

FRED-QD is, however, a U.S. database. Its 245 series are drawn from FRED, which in turn mirrors U.S. statistical agencies (the Bureau of Economic Analysis, the Bureau of Labor Statistics, the Census Bureau, the Federal Reserve). A researcher wanting the same kind of standardized, quarterly, factor-model-ready panel for a single country other than the United States has, since 2024, had EA-MD-QD to draw an Austrian sub-panel from – but only as a byproduct of a euro-area-wide construction, at a scale (1,136 series, distributed as spreadsheets plus MATLAB code) oriented toward cross-country panel analysis rather than a single country's own use.

This report's contribution is accordingly framed as a *methodology* rather than a competing dataset: a construction and verification procedure that, applied to Austria as a reference case, produces a panel parameterized by country rather than hard-coded (the underlying tool already has built-in country-code support for 38 OECD member states), and that trades EA-MD-QD's scale for two properties a country-first, code-based construction makes easier to guarantee. First, every series' exact source and any conceptual caveat is recorded in a public, machine-readable registry rather than left to spreadsheet documentation (Section 3). Second, the fetch pipeline itself is open and re-runnable rather than a periodically-refreshed static release, so a researcher can audit, correct, or extend any single series' provenance directly, and repeat the same procedure for a country neither FRED-QD nor EA-MD-QD covers. Rather than replicate FRED-QD's exact 245 series – most of which, as documented in Section 4, have no cross-country equivalent at all – it reproduces FRED-QD's *design*: one representative concept per group, sourced from the most authoritative and freshest agency available, with every source verified against its live API rather than assumed correct from documentation alone.

The remainder of this report is organized as follows. Section 2 describes how the panel is constructed: the concept taxonomy, the source hierarchy and the country-specific overrides that take precedence over it, the verification methodology, and the update cadence. Section 3 describes the data-sources registry itself. Section 4 situates the 38 implemented concepts against FRED-QD's full 245-series catalog. Section 5 documents known data-quality issues and gaps, following FRED-QD's own convention of disclosing rather than papering over them. Section 6 outlines concrete extensions. Section 7 concludes. Appendix A reproduces the full FRED-QD variable catalog; Appendix B gives the current Austria-specific source mapping.

2 Data Construction

2.1 Concept taxonomy and canonical schema

Every country the underlying tool (`scripts/build_country_panel.R`) is run for produces a panel with exactly the same 38 columns, in the same order, regardless of which concepts actually resolved for that country: a concept with no available source for a given country is still present as an all-`NA` column rather than silently missing. This mirrors, in spirit, FRED-QD's own commitment to a fixed, documented series list – and is a deliberate design choice separate from mirroring FRED-QD's specific series: the point of routing every country through the same 25 FRED-QD-style concept labels is that switching the country argument should be the *only* thing that changes between two runs, so that downstream code can load any country's file with identical column-handling logic. Table 1 lists the 38 concepts, the FRED-QD group each belongs to, and – alongside the FRED-QD mnemonic each already approximates for the United States – the corresponding quarterly series ID from EA-MD-QD (Barigozzi et al., 2026), the large existing euro-area dataset already discussed in Section 6, where one exists. This puts the US reference, the EA reference and this project's own Austrian construction of the same concept side by side in one place; Appendix B gives the exact Austrian source and any conceptual caveat.

FRED-QD Group	**Concept**	**FRED-QD ID (US)**	**EA-MD-QD ID (EA)**
Output and Income	`real_gdp`	`GDPC1`	`GDP`
Output and Income	`real_household_consumption`	`PCECC96`	`HFCE`
Output and Income	`real_govt_consumption`	`GCEC1`	`GFCE`
Output and Income	`real_gfcf_total`	`FPIx`	`GCFC`
Output and Income	`real_exports`	`EXPGSC1`	`EXPGS`
Output and Income	`real_imports`	`IMPGSC1`	`IMPGS`
Output and Income	`real_household_disposable_income`	`DPIC96`	`AHRDI`[†]
Industrial Production	`industrial_production`	`INDPRO`	*none*
Industrial Production	`industrial_confidence`	*none*	*none*
Employment and Unemployment	`unemployment_rate`	`UNRATE`	*none*
Employment and Unemployment	`employment_rate`	*none*	*none*
Employment and Unemployment	`employment_expectations`	*none*	*none*
Housing	`house_price_real`	`USSTHPI`	`HPRC`
Housing	`construction_confidence`	*none*	*none*
Inventories, Orders, and Sales	`retail_sales_volume`	`RSAFSx`	*none*
Inventories, Orders, and Sales	`retail_confidence`	*none*	*none*
Prices	`cpi_index`	`CPIAUCSL`	*none*
Prices	`core_cpi_index`	`CPILFESL`	*none*
Prices	`food_price_index`	*none*	*none*
Prices	`energy_price_index`	*none*	*none*
Prices	`services_price_index`	`CUSR0000SAS`	*none*
Earnings and Productivity	`unit_labor_cost`	`ULCNFB`	*none*
Interest Rates	`long_term_rate`	`GS10`	*none*
Interest Rates	`short_term_rate`	`TB3MS`	*none*

FRED-QD Group	Concept	FRED-QD ID (US)	EA-MD-QD ID (EA)
Interest Rates	`mortgage_rate`	`MORTGAGE30US`	*none*
Money and Credit	`credit_to_private_nonfin_sector`	*none*	*none*
Money and Credit	`household_mortgage_loans`	`REALLNx`	`HHLB.LLN`*
Household Balance Sheets	`euro_area_household_net_worth_growth`	`TNWBSHNOx`	*none*
Household Balance Sheets	`household_credit_to_gdp`	*none*	`HHLB`*
Non-Household Balance Sheets	`corporate_credit_to_gdp`	*none*	`NFCLB`*
Non-Household Balance Sheets	`government_debt_to_gdp`	`GFDEGDQ188S`	`GGLB`*
Exchange Rates	`fx_rate_to_usd`	*none*	*none*
Exchange Rates	`real_effective_exchange_rate`	`TWEXAFEGSMTHx`	*none*
Other	`consumer_confidence`	`UMCSENTx`	*none*
Other	`economic_sentiment_indicator`	*none*	*none*
Other	`services_confidence`	*none*	*none*
Other	`geopolitical_risk`	*none*	*none*
Stock Markets	`share_price_index`	`S&P 500`	*none*

FRED-QD Group	Concept	FRED-QD ID (US)	EA-MD-QD ID (EA)

Table 1: The 38 implemented concepts, their FRED-QD group, and their US (FRED-QD) and EA (EA-MD-QD, quarterly series only) cross-references. Fourteen concepts have no direct FRED-QD mnemonic: five standard cross-country indicators FRED-QD has no equivalent for at all (employment rate, credit to the private non-financial sector, household credit-to-GDP, corporate credit-to-GDP, the FX rate to USD), two HICP sub-categories with no standalone FRED-QD counterpart (food, energy), and seven survey- or research-index-based concepts entirely outside FRED-QD's scope (economic sentiment, industrial/services/retail/construction confidence, employment expectations, geopolitical risk) – see Section 2.2 for the seven's sourcing. Twelve concepts have an EA-MD-QD quarterly cross-reference: seven are exact matches (same concept, same construction); four (marked *) are the closest EA-MD-QD counterpart to a BIS-sourced %-of-GDP ratio this project publishes, but are themselves raw ECB/Eurostat Quarterly Sector Accounts currency LEVELS, not ratios – related, not identical; one (marked †, household disposable income) is a series EA-MD-QD itself confirms has no observations for Austria, independently corroborating the same gap Section 5 already documents. The remaining 26 concepts have no quarterly EA-MD-QD counterpart: most of EA-MD-QD's own series for these concepts exist only at monthly frequency (all its Interest Rates, Financial Markets, Industrial Production and Turnover, Confidence Indicators, and Prices series other than the GDP deflator), and a few concepts – geopolitical risk chief among them – fall outside EA-MD-QD's scope entirely.

2.2 Source hierarchy and country-specific overrides

For the seven "anchor" national-accounts concepts (real GDP, household consumption, government consumption, gross fixed capital formation, exports, imports, and household disposable income), the tool queries both Eurostat's Quarterly National Accounts (`namq_10_gdp`) for EU member states and the OECD's Quarterly National Accounts (`DF_QNA`), and merges the two rather than treating either as a pure fallback for the other: Eurostat's value is kept at every period it covers, and OECD's value fills in any period Eurostat does not, with the IMF's National Economic Accounts (`QNEA`) as a final fallback for whatever neither source has at all. Eurostat is preferred where both cover the same period not merely because it is the more "local" source, but because in at least one case it is conceptually closer to FRED-QD's own definition: Eurostat's household-consumption transaction (`P31_S14`) covers households only, while the equivalent OECD sector (`S1M`) also includes non-profit institutions serving households, making it broader than FRED-QD's household-only personal consumption expenditure. Merging rather than strictly falling back matters because the two sources' histories differ substantially: Eurostat's own Austrian series start at 1995-Q1 for every anchor concept (confirmed live – querying as far back as 1970-Q1 returns nothing earlier), while OECD QNA's Austrian real GDP series was confirmed live to extend to 1960-Q1. Treating OECD as a fallback used only when Eurostat returns nothing at all would have left 35 years of available history unused for every concept Eurostat covers even partially; the merge recovers it, extending this project's default starting point from 1995-Q1 to 1960-Q1 – four decades earlier than EA-MD-QD's stated January 2000 starting vintage (Barigozzi and Lissona, 2024).

Merging two sources is not simply coalescing them, however: a plain period-by-period coalesce (`merge_prefer()` in the code) silently assumes both sources report the SAME quantity on the SAME scale wherever one takes over from the other, an assumption this project's own plausibility checks (Section 2.6) found to be false here. OECD's QNA table (`T0102`, used for every anchor concept) only offers `TRANSFORMATION="LA"` ("Annual levels", i.e. the quarterly series expressed at an annualized rate) – confirmed live by querying `TRANSFORMATION="N"` ("Non transformed data") and receiving a clean `NoRecordsFound` for both a 1990s and a 2020s period, not a guess – while Eurostat reports true, non-annualized quarterly levels. Coalescing the two as-is therefore introduced a ∼75% level discontinuity at the exact quarter each anchor concept's source switches from OECD to Eurostat (1994-Q4 to 1995-Q1 for Austria), across all six concepts alike. The fix, `splice_prefer()`, rescales OECD's contribution by the ratio of the two sources' values at their one genuine overlap point (OECD is queried for the full requested range regardless of where Eurostat takes over, so this overlap already exists in the fetched data) before coalescing – preserving OECD's own internally-consistent quarter-to- quarter dynamics for the period Eurostat does not cover, while correcting its absolute level to match Eurostat's. Section 2.6 describes how this was found.

For credit and balance-sheet concepts, the BIS's Credit to the Non-Financial Sector dataflow (`WS_TC`) is queried once per borrower sector (private non-financial sector, households, non-financial corporations, and general government) for *all* countries at once – a wildcarded `BORROWERS_CTY` dimension was confirmed live to return every country BIS covers in a single request – and the result cached locally so that building panels for multiple countries in one session does not re-download the same data. The general-government sector doubles as this project's source for `government_debt_to_gdp`: BIS's own credit-statistics methodology treats "credit to general government" as a close proxy for gross government debt, confirmed live for Austria, Germany and the United States alike (75–111% of GDP across the three as of 2025-Q4) – a genuinely cross-country source, unlike every other override in this section, none of which extend outside the EU or euro area. The ECB's Quarterly Sector Accounts supply household net worth, but only as a euro-area *aggregate*: the dataflow was confirmed, by enumerating its actual published series keys rather than assuming per-country availability, to have observations only for the fixed-composition euro-area total, not for individual member states. The ECB's MFI Interest Rate Statistics, by contrast, are genuinely country-specific, and supply a mortgage rate (new-business loans to households for house purchase) for any euro-area member. A second ECB dataflow, MFI Balance Sheet Items (BSI), supplies the natural counterpart to that rate: the outstanding stock of the same loan category (`household_mortgage_loans`), confirmed current for both Austria (EUR 133.0 billion) and Germany (EUR 1,658.4 billion) as of July 2026. BSI and MIR are both ECB MFI statistics about the same underlying loans, but use entirely different dimension structures and codes – constructing the BSI key by analogy with MIR's (a reasonable-looking guess) returned a structurally valid zero-observation response even for the simplest total-loans case; the working key was instead found by searching the ECB Data Portal's own published series list for its human-readable title, which exposes the exact SDMX key alongside it.

For the remaining concepts (industrial production, unemployment, house prices, consumer prices, consumer sentiment, share prices, and several others), the default source is FRED's own public mirror of OECD Main Economic Indicators and BIS series – reached via the same `fredgraph.csv` export used throughout FRED-QD's own construction, requiring no API key. Four deliberate exceptions take precedence over this default. Two are staleness fixes, each because the FRED-mirrored OECD series was confirmed live to have stopped updating (frozen since 2023–2024, depending on the series and country) while a fresher EU primary source exists: for EU member states, consumer confidence is sourced from the European Commission's own Business and Consumer Survey; and, for the same EU member states, the consumer price index is sourced from Eurostat's own Harmonised Index of Consumer Prices (`prc_hicp_midx`), confirmed live to extend roughly two years further (to 2025-Q4 for Austria) than the frozen FRED mirror (2023-Q4). A third is a conceptual-fidelity fix rather than a staleness one: for EU member states where Eurostat publishes an index-level series for it, unit labor cost is sourced from Eurostat's labour productivity and unit-labour-cost dataflow (`namq_10_lp_ulc`, `NA_ITEM= NULC_HW`), which is hours-based like FRED-QD's own `ULCNFB` construction, unlike the FRED-mirrored OECD proxy's employment-based percentage change – confirmed available in this index-level form for Austria but, on the identical live query, confirmed *absent* for Germany (only percentage-change variants are published there), which keeps the FRED-mirror value. This is the one override in this project that can genuinely fail for an EU member country, not only outside the EU – documented here rather than assumed to generalize from the Austrian case it was verified against. The fourth exception is Austria-specific: the share-price-index concept is sourced from the ATX (Austria's own benchmark index) via Yahoo Finance, in place of a generic OECD "all shares" proxy.

Beyond these four exceptions to an existing default, the Prices group also gains four concepts with *no* FRED-mirror default to override at all: the HICP fetcher above generalizes to any COICOP category, so the same verified dataflow and key structure also supply core inflation (`TOT_X_NRG_FOOD`, excluding energy and food – the standard ECB/Eurostat measure, and FRED-QD's closest analog to `CPILFESL`), food (`CP01`), energy (`NRG`), and services (`SERV`), all confirmed live for both Austria and Germany. These are new `core_cpi_index/food_price_index/energy_price_index/services_price_index` concepts, EU-only by construction, since no international default exists for them to fall back to outside the EU.

The same archive already fetched for consumer confidence carries six further per-country columns this project did not previously use – confirmed live by enumerating the cached workbook's own header row, not assumed from the source's documentation. Two were prioritized for their documented predictive value rather than mere availability: the Economic Sentiment Indicator (`economic_sentiment_indicator`, column suffix `ESI`) is DG ECFIN's own flagship composite – a weighted average of the industry, services, consumer, retail and construction survey balances – explicitly constructed and empirically validated to track and lead euro-area GDP growth; the Industrial Confidence Indicator (`industrial_confidence`, `INDU`) is one of the archive's oldest series (published since 1985) and a standard input to the OECD's own Composite Leading Indicators for many countries. A third, `employment_expectations` (`EEI`), is DG ECFIN's own purpose-built leading indicator for employment turning points, introduced in 2013 specifically because the sectoral surveys' employment sub-components lead employment growth. The remaining three – services, retail and construction confidence (`SERV/ RETA/BUIL`) – are the archive's other ESI sub-components: standard EC-published sentiment measures without the same individually-validated leading-indicator literature behind ESI, INDU or EEI specifically, but a low-cost addition once the underlying fetcher was already generalized to accept any of the seven column suffixes.

Geopolitical uncertainty is sourced from the Geopolitical Risk (GPR) index of Caldara and Iacoviello (2022), the standard academic and policy measure of adverse geopolitical events and their associated risks, downloaded directly from the authors' own published monthly data file rather than reimplemented. The source constructs country-specific indices for 44 countries, confirmed by enumerating the file's own column names rather than assumed from its documentation – a check that also surfaced a genuine trap: the file's "GPRC_AUS" column is Australia, not Austria, the same two-versus-three-letter country-code collision this project's own country-code table exists to prevent for FRED's mirror series. Austria is confirmed *absent* from the 44 country-specific series; `geopolitical_risk` falls back to the source's global index for Austria and any other country without its own column, rather than leaving the concept unresolved, since the index's own validation literature documents international spillovers of geopolitical risk shocks independent of a country's own media coverage of them. Germany and the United States do have their own country-specific series, confirmed live and current through July 2026.

2.3 Verification methodology

Every source used in this project was confirmed against its live API – structure and codelist queries where relevant, then a real data pull returning current, plausible values for at least two countries – before being wired into the panel-construction tool. This project's predecessor script guessed several dataflow identifiers and dimension orderings that later verification found to be wrong (documented in each module's header comments, e.g. `R/oecd.R`, `R/bis.R`, `R/ecb.R`); the verification pass this report describes was undertaken specifically to replace those guesses with confirmed keys, correcting several outright errors along the way. Series accepted as "verified" report a real non-empty response with plausible values, not merely an HTTP 200 status, since several sources return a syntactically valid but semantically empty or stale response for an unsupported query – for example, an unpublished monthly archive redirecting to a generic landing page rather than returning a 404.

2.4 Update cadence

Following FRED-QD's own practice of publishing a new vintage on the last business day of each month, a scheduled job rebuilds the Austrian (and two comparison, German and U.S.) panels on the first of each month and archives a dated copy of both the data and its coverage report into `output/vintages/`, so that a given month's release remains reproducible even after later revisions are folded into the "current" file – the same rationale FRED-QD gives for retaining its own historical vintages.

2.5 Transformation codes

Where a concept corresponds to an actual FRED-QD series, this project reuses that series' published transformation code (Appendix A) rather than re-deriving one, for the same reason FRED-QD itself reuses Stock and Watson (2012)'s original codes where available: consistency with the literature that already uses these codes, rather than introducing an independent judgment call about each series' order of integration.

2.6 Plausibility checks: verification without a ground truth

Section 2.2 above (*Verification methodology*) describes how every SOURCE was confirmed against its live API before being wired into the tool. That check happens once, when a module is built. A separate question is whether the OUTPUT a given run actually produces still looks right – and for the United States, § 4's `--validate` flag answers exactly that, by correlating this project's growth rates against the real published FRED-QD file. No such file exists for any other country, including Austria itself; without a second check, a researcher extending this project to a new country would have no automated signal at all pointing at which of the panel's 38 columns might be wrong, only the option of reading every value by hand.

`R/plausibility_checks.R` closes that gap with checks that need no ground truth, only the panel itself and each concept's known measurement type (percentage/ratio, survey balance or sentiment index, quarterly growth rate, or level/index): a broad but real-world-informed range for the first three types, and, for levels and indices, both positivity and the absence of an implausible quarter-over-quarter jump (a heuristic threshold, loose enough to tolerate genuine volatility – confirmed live not to flag real crisis-era spikes in Austria's own `geopolitical_risk` series, which move far more than any whole-economy aggregate around events like the Gulf War or Russia's 2022 invasion of Ukraine). These are deliberately loose bounds, not authoritative thresholds: the goal is a prioritized list of which concepts most reward a researcher's limited manual-verification time, not a certificate of correctness. Results are written into `<country>_coverage.json` alongside the existing resolved/skipped breakdown, and summarized on the console after every run.

This is not a hypothetical safeguard: on its first live run, it caught a real, previously-unknown bug (the OECD/Eurostat level-splice issue described in Section 2.2) that had been present, and invisible to `--validate`, since the anchor-concept merge was introduced. It also surfaced a second, deeper finding not yet acted on: `cpi_index`'s FRED-mirror default (used for non-EU countries) is itself a percentage-change series, not the index level its own FRED-QD mnemonic (`CPIAUCSL`) implies – flagged as a concrete first task for a researcher extending this project, in `CONTRIBUTING.md`.

3 The Data-Sources Registry

`docs/data_sources.csv` is a long-format table – one row per (country, concept) pair – with columns `country`, `variable`, `provider`, `key`, and `comment`. It is the single documentation surface for "what source backs this number, and how does it conceptually differ from the US/FRED-QD definition," replacing what would otherwise be prose comments scattered across each source-specific module. The `USA` rows are the deliberate exception: they always hold the actual FRED-QD mnemonic for each concept, since FRED-QD itself is the ground truth every other country's row approximates, and are not overwritten by a live run through the international sources for the United States. Appendix B reproduces the current Austria rows.

4 Coverage Relative to FRED-QD

Of FRED-QD's 245 series, 38 concepts are currently implemented for Austria (24 correspond directly to a specific FRED-QD mnemonic; the remaining fourteen have none, as Table 1's caption details – five standard cross-country indicators, two HICP sub-categories, and seven survey- or research-index-based concepts sourced from the European Commission's own survey archive and an external academic index, respectively, discussed in Section 2.2). A companion file, `docs/candidate_indicators_austria.csv`, goes through every one of the remaining 207 FRED-QD series individually and classifies each as a CANDIDATE (a proposed but unverified Eurostat/ECB/OECD source, with a confidence rating), DERIVABLE (computable from concepts already implemented, e.g. export and import shares of GDP), or NO_EQUIVALENT (a genuinely U.S.-specific construct, with the reason stated – for example, the federal/state/local government employment split has no counterpart in European labour-market statistics). Unlike the registry described in Section 3, that file is an explicitly unverified research proposal, intended as a prioritized starting point for future verification passes rather than a claim that any given source is confirmed to work.

5 Known Limitations

Following FRED-QD's own convention of documenting data-quality issues rather than silently working around them, several are noted here explicitly. First, the OECD-mirrored consumer price index was found, on live inspection, to have stopped updating at 2023-Q4 for Austria – the same underlying FRED-mirror family already fixed for consumer confidence via the EC survey override. For EU member states this is now fixed the same way, via an override to Eurostat's own HICP dataflow (see Section 2.2), confirmed live to extend coverage to 2025-Q4; non-EU countries (including the United States) still receive the stale FRED-mirror value, since no equivalent EU-style primary source exists for them. Second, OECD's own data API was found to rate-limit under the moderate request volume of building three countries' panels back-to-back; the IMF fallback absorbs this in practice; but it means a much larger cross-country loop should expect to encounter it. Third, household net worth is available only as a euro-area aggregate, not per country, a genuine data availability limit rather than an implementation gap. Fourth, the unit labor cost override is the one exception to the pattern that a source either works for every country in its stated scope (EU, euro area) or none: Eurostat's index-level series for it is confirmed present for Austria but confirmed absent for Germany, an EU member, even though both are otherwise EU-anchor-concept countries in this project – future countries added to this project should not assume EU membership alone predicts whether this particular override will apply. Fifth, several concepts are structurally unavailable outside the euro area or the EU respectively, so a non-EU country's panel will show these as `NA` by construction, not by omission: mortgage rate and household mortgage loans (euro area), and the four HICP sub-categories plus six of the seven EC Business and Consumer Survey concepts (all but consumer confidence, which alone has a FRED-mirror fallback that also works outside the EU) for the EU. Combined with the pre-existing genuine data gaps this section already documents – retail sales volume specifically for the US, the euro-area-only household-net-worth aggregate, and the FX-rate concept that is not meaningful for the US itself – this means sixteen of the panel's 38 concepts are necessarily `NA` for the United States, not a coverage failure of this project's own sources. The one exception to this section's pattern of EU/euro-area-only overrides is geopolitical risk, which – unlike every other addition here – resolves for every country regardless of EU membership (Section 2.2). Sixth, the anchor-concept merge itself had a genuine level-scale bug – found and fixed the same day by the plausibility checks introduced in this pass (Section 2.6); see Section 2.2 for the full account. It is recorded here, not just where it was fixed, because FRED-QD's own convention this section follows is to document data-quality issues rather than let a fix quietly erase the fact that a real error existed in a previously-distributed vintage of this dataset. Seventh, and NOT yet fixed: `cpi_index`'s FRED-mirror default for non-EU countries was found, by the same plausibility checks, to be a percentage-change series rather than the index level its own FRED-QD mnemonic implies (Section 2.6) – flagged as a concrete first task in `CONTRIBUTING.md` rather than guessed at under this report's own writing deadline. Finally, quarterly household disposable income was not found for Austria, Germany, or the United States themselves through either OECD or Eurostat, consistent with McCracken and Ng's own observation that FRED-QD's income-side detail is harder to source than its expenditure side.

6 Future Work

The candidate list in Section 4 identifies several concrete, high-confidence extensions: Eurostat's Harmonised Index of Consumer Prices, already implemented for the overall `cpi_index` concept plus core, food, energy and services (Section 2.2), could, via its finer COICOP breakdown, also supply Austrian counterparts to most of FRED-QD's remaining PCE/CPI sub-category price indices (apparel, transport, health, and the various all-items-less-X variants); Eurostat's labour productivity and unit-labour-cost dataflow (`namq_10_lp_ulc`), already used for the whole-economy `unit_labor_cost` override (Section 2.2), also carries sector-specific output-per-hour and unit-labor-cost series that would be a direct source for FRED-QD's `OPHMFG/OPHNFB/ OPHPBS/ULCBS/ULCMFG` candidates, none of which this project currently tracks; the ECB's MFI Balance Sheet Items, already used for `household_mortgage_loans` (Section 2.2), carries the same loan-by-purpose breakdown for other counterpart sectors and purposes – `BS_ITEM=A21T` ("Credit for consumption"), by direct analogy with the `A22T` ("Lending for house purchase") code already confirmed working, is the natural next candidate for `CONSUMERx`; and Eurostat's quarterly government debt (Maastricht debt) series remains a candidate for `GFDEBTNx`, the real-dollar-level federal-debt concept BIS does not publish (its %-of-GDP counterpart, `GFDEGDQ188S`, is already implemented via BIS – Section 2.2). The European Commission's AMECO database was evaluated and deliberately not integrated into the quarterly panel, since it is annual only; it remains a candidate for a separate annual companion file covering fiscal concepts absent from both FRED-QD and this project's current scope.

A further extension comes from cross-checking the candidate list against EA-MD-QD's own published data description (Barigozzi et al., 2026), which – being already live, verified, and Austria-specific for every series it carries – serves as independent corroboration wherever its coverage overlaps this project's candidates. Two of this report's lowest-confidence candidates turn out to be directly confirmed: EA-MD-QD's Table 1 lists Austria-specific quarterly household total financial assets and liabilities (`HHASS`, `HHLB`, sourced from the same Eurostat/ECB Quarterly Sector Accounts family as this project's euro-area-only household net worth series) and the analogous non-financial-corporation aggregates (`NFCASS`, `NFCLB`) as genuinely available per country, not merely euro-area-wide – upgrading four Household and Non-Household Balance Sheet candidates from an unconfirmed LOW to a verified-elsewhere HIGH in the updated `docs/candidate_indicators_austria.csv`. EA-MD-QD's variable list also covers several concepts with no FRED-QD counterpart at all – a household gross saving rate, household and corporate investment and profit shares, a semi-durable-goods consumption category sitting between FRED-QD's durable and non-durable splits, a real effective exchange rate, and per-worker (rather than per-hour) labour productivity. A seventh, sector-specific confidence balances (construction, retail, services) obtainable from the same European Commission survey archive this project already fetches for consumer confidence, has since been implemented (Section 2.2), alongside two further survey concepts (economic sentiment, employment expectations) EA-MD-QD's own list does not carry. The remaining six concepts are catalogued separately in `docs/candidate_indicators_ea_md_qd.csv`, alongside FRED-QD's own list, rather than folded into Appendix B's mnemonic-keyed table, since they fall outside FRED-QD's 245-series scope by construction. Finally, in the spirit of McCracken and Ng (2020)'s own empirical validation of FRED-QD via factor estimation and forecasting exercises, a natural extension once coverage and country count grow further is to assess whether factors extracted from this panel behave comparably to those extracted from FRED-QD itself over the sample period they overlap.

7 Conclusion

This report's contribution is the construction and verification methodology it describes, not the size of the panel it happens to produce: a fixed concept taxonomy, a source hierarchy with explicit country-specific overrides, every source confirmed against its live API rather than assumed correct, a public registry of exactly what was used and why, and an explicit accounting of what remains uncovered and why. Austria serves here as the reference case precisely because a larger-scale alternative already exists for it (EA-MD-QD, Barigozzi and Lissona, 2024) – so the value on offer is not a bigger panel, but a smaller, fully transparent, and openly re-runnable one that the same procedure can reproduce for any other country, together with a concrete, prioritized, and honestly-labeled path to extending its coverage further.

References

Barigozzi, M. and Lissona, C. (2024). EA-MD-QD: Large Euro Area and Euro Member Countries Datasets for Macroeconomic Research. *Zenodo dataset*, `https://doi.org/10.5281/zenodo.10514667`.

Barigozzi, M., Lissona, C., and Tonni, L. (2026). Large datasets for the euro area and its member countries and the dynamic effects of the common monetary policy. *arXiv preprint*, arXiv:2410.05082.

Caldara, D. and Iacoviello, M. (2022). Measuring Geopolitical Risk. *American Economic Review*, 112(4):1194–1225.

Fortin-Gagnon, O., Leroux, M., Stevanovic, D., and Surprenant, S. (2018). A Large Canadian Database for Macroeconomic Analysis. *CIRANO Working Papers*, 2018s-25.

McCracken, M. W. and Ng, S. (2016). FRED-MD: A Monthly Database for Macroeconomic Research. *Journal of Business & Economic Statistics*, 34(4):574–589.

McCracken, M. W. and Ng, S. (2020). FRED-QD: A Quarterly Database for Macroeconomic Research. *Federal Reserve Bank of St. Louis Working Paper*, 2020-005. `https://doi.org/10.20955/wp.2020.005`.

Stock, J. H. and Watson, M. W. (2012). Disentangling the Channels of the 2007–2009 Recession. *NBER Working Paper*, 18094.

A FRED-QD Variable Catalog

For reference, Table 1 lists this project's 38 implemented concepts and their FRED-QD group; the tables that follow reproduce FRED-QD's complete 245-series catalog (McCracken and Ng, 2020), grouped into the original fourteen categories, with each series' recommended transformation code and whether Stock and Watson (2012) used it when estimating factors (✓ = yes). If the FRED mnemonic does not end in "x" the series comes directly from the FRED database; otherwise it is a modified variant, most commonly a nominal series manually deflated by the PCE or core-PCE price index. Monthly-frequency source series are aggregated to quarterly frequency by averaging over the three months of the quarter.

Group: Output and Income

Mnemonic	Description	Tcode	SW
GDPC1	Real Gross Domestic Product, 3 Decimal (Billions of Chained 2012 Dollars)	5	
PCECC96	Real Personal Consumption Expenditures (Billions of Chained 2012 Dollars)	5	
PCDGx	Real Personal Consumption Expenditures: Durable Goods (Billions of Chained 2012 Dollars), deflated using PCE	5	✓
PCESVx	Real Personal Consumption Expenditures: Services (Billions of 2012 Dollars), deflated using PCE	5	✓
PCNDx	Real Personal Consumption Expenditures: Nondurable Goods (Billions of 2012 Dollars), deflated using PCE	5	✓
GPDIC1	Real Gross Private Domestic Investment, 3 Decimal (Billions of Chained 2012 Dollars)	5	
FPIx	Real Private Fixed Investment (Billions of Chained 2012 Dollars), deflated using PCE	5	
Y033RC1Q027SBEAx	Real Gross Private Domestic Investment: Fixed Investment: Nonresidential: Equipment (Billions of Chained 2012 Dollars), deflated using PCE	5	✓
PNFIx	Real Private Fixed Investment: Nonresidential (Billions of Chained 2012 Dollars), deflated using PCE	5	✓
PRFIx	Real Private Fixed Investment: Residential (Billions of Chained 2012 Dollars), deflated using PCE	5	✓
A014RE1Q156NBEA	Shares of Gross Domestic Product: Gross Private Domestic Investment: Change in Private Inventories (Percent)	1	✓
GCEC1	Real Government Consumption Expenditures & Gross Investment (Billions of Chained 2012 Dollars)	5	

Mnemonic	Description	Tcode	SW
A823RL1Q225SBEA	Real Government Consumption Expenditures and Gross Investment: Federal (Percent Change from Preceding Period)	1	✓
FGRECPTx	Real Federal Government Current Receipts (Billions of Chained 2012 Dollars), deflated using PCE	5	✓
SLCEx	Real Government State and Local Consumption Expenditures (Billions of Chained 2012 Dollars), deflated using PCE	5	✓
EXPGSC1	Real Exports of Goods & Services, 3 Decimal (Billions of Chained 2012 Dollars)	5	✓
IMPGSC1	Real Imports of Goods & Services, 3 Decimal (Billions of Chained 2012 Dollars)	5	✓
DPIC96	Real Disposable Personal Income (Billions of Chained 2012 Dollars)	5	
OUTNFB	Nonfarm Business Sector: Real Output (Index 2012=100)	5	
OUTBS	Business Sector: Real Output (Index 2012=100)	5	
OUTMS	Manufacturing Sector: Real Output (Index 2012=100)	5	
B020RE1Q156NBEA	Shares of Gross Domestic Product: Exports of Goods and Services (Percent)	2	
B021RE1Q156NBEA	Shares of Gross Domestic Product: Imports of Goods and Services (Percent)	2	

Group: Industrial Production

Mnemonic	Description	Tcode	SW
INDPRO	Industrial Production Index (Index 2012=100)	5	
IPFINAL	Industrial Production: Final Products (Market Group) (Index 2012=100)	5	
IPCONGD	Industrial Production: Consumer Goods (Index 2012=100)	5	
IPMAT	Industrial Production: Materials (Index 2012=100)	5	
IPDMAT	Industrial Production: Durable Materials (Index 2012=100)	5	✓
IPNMAT	Industrial Production: Nondurable Materials (Index 2012=100)	5	✓
IPDCONGD	Industrial Production: Durable Consumer Goods (Index 2012=100)	5	✓
IPB51110SQ	Industrial Production: Durable Goods: Automotive Products (Index 2012=100)	5	✓
IPNCONGD	Industrial Production: Nondurable Consumer Goods (Index 2012=100)	5	✓
IPBUSEQ	Industrial Production: Business Equipment (Index 2012=100)	5	✓
IPB51220SQ	Industrial Production: Consumer Energy Products (Index 2012=100)	5	✓
TCU	Capacity Utilization: Total Industry (Percent of Capacity)	1	✓

Mnemonic	Description	Tcode	SW
CUMFNS	Capacity Utilization: Manufacturing (SIC) (Percent of Capacity)	1	✓
IPMANSICS	Industrial Production: Manufacturing (SIC) (Index 2012=100)	5	
IPB51222S	Industrial Production: Residential Utilities (Index 2012=100)	5	
IPFUELS	Industrial Production: Fuels (Index 2012=100)	5	

Group: Employment and Unemployment

Mnemonic	Description	Tcode	SW
PAYEMS	All Employees: Total Nonfarm (Thousands of Persons)	5	
USPRIV	All Employees: Total Private Industries (Thousands of Persons)	5	
MANEMP	All Employees: Manufacturing (Thousands of Persons)	5	
SRVPRD	All Employees: Service-Providing Industries (Thousands of Persons)	5	
USGOOD	All Employees: Goods-Producing Industries (Thousands of Persons)	5	
DMANEMP	All Employees: Durable Goods (Thousands of Persons)	5	✓
NDMANEMP	All Employees: Nondurable Goods (Thousands of Persons)	5	
USCONS	All Employees: Construction (Thousands of Persons)	5	✓
USEHS	All Employees: Education & Health Services (Thousands of Persons)	5	✓
USFIRE	All Employees: Financial Activities (Thousands of Persons)	5	✓
USINFO	All Employees: Information Services (Thousands of Persons)	5	✓
USPBS	All Employees: Professional & Business Services (Thousands of Persons)	5	✓
USLAH	All Employees: Leisure & Hospitality (Thousands of Persons)	5	✓
USSERV	All Employees: Other Services (Thousands of Persons)	5	✓
USMINE	All Employees: Mining and Logging (Thousands of Persons)	5	✓
USTPU	All Employees: Trade, Transportation & Utilities (Thousands of Persons)	5	✓
USGOVT	All Employees: Government (Thousands of Persons)	5	
USTRADE	All Employees: Retail Trade (Thousands of Persons)	5	✓
USWTRADE	All Employees: Wholesale Trade (Thousands of Persons)	5	✓
CES9091000001	All Employees: Government: Federal (Thousands of Persons)	5	✓
CES9092000001	All Employees: Government: State Government (Thousands of Persons)	5	✓
CES9093000001	All Employees: Government: Local Government (Thousands of Persons)	5	✓

Mnemonic	Description	Tcode	SW
CE16OV	Civilian Employment (Thousands of Persons)	5	
CIVPART	Civilian Labor Force Participation Rate (Percent)	2	
UNRATE	Civilian Unemployment Rate (Percent)	2	
UNRATESTx	Unemployment Rate: Less than 27 Weeks (Percent)	2	
UNRATELTx	Unemployment Rate: More than 27 Weeks (Percent)	2	
LNS14000012	Unemployment Rate: 16 to 19 Years (Percent)	2	✓
LNS14000025	Unemployment Rate: 20 Years and Over, Men (Percent)	2	✓
LNS14000026	Unemployment Rate: 20 Years and Over, Women (Percent)	2	✓
UEMPLT5	Number of Civilians Unemployed: Less than 5 Weeks (Thousands of Persons)	5	✓
UEMP5TO14	Number of Civilians Unemployed for 5 to 14 Weeks (Thousands of Persons)	5	✓
UEMP15T26	Number of Civilians Unemployed for 15 to 26 Weeks (Thousands of Persons)	5	✓
UEMP27OV	Number of Civilians Unemployed for 27 Weeks and Over (Thousands of Persons)	5	✓
LNS13023621	Unemployment Level: Job Losers (Thousands of Persons)	5	✓
LNS13023557	Unemployment Level: Reentrants to Labor Force (Thousands of Persons)	5	✓
LNS13023705	Unemployment Level: Job Leavers (Thousands of Persons)	5	✓
LNS13023569	Unemployment Level: New Entrants (Thousands of Persons)	5	✓
LNS12032194	Employment Level: Part-Time for Economic Reasons, All Industries (Thousands of Persons)	5	✓
HOABS	Business Sector: Hours of All Persons (Index 2012=100)	5	
HOAMS	Manufacturing Sector: Hours of All Persons (Index 2012=100)	5	
HOANBS	Nonfarm Business Sector: Hours of All Persons (Index 2012=100)	5	
AWHMAN	Average Weekly Hours of Production and Nonsupervisory Employees: Manufacturing (Hours)	1	✓
AWHNONAG	Average Weekly Hours of Production and Nonsupervisory Employees: Total Private (Hours)	2	✓
AWOTMAN	Average Weekly Overtime Hours of Production and Nonsupervisory Employees: Manufacturing (Hours)	2	✓
HWIx	Help-Wanted Index	1	
UEMPMEAN	Average (Mean) Duration of Unemployment (Weeks)	2	
CES0600000007	Average Weekly Hours of Production and Nonsupervisory Employees: Goods-Producing (Hours)	2	
HWIURATIOx	Ratio of Help Wanted to Number Unemployed	2	

Mnemonic	Description	Tcode	SW
CLAIMSx	Initial Claims	5	

Group: Housing

Mnemonic	Description	Tcode	SW
HOUST	Housing Starts: Total: New Privately Owned Housing Units Started (Thousands of Units)	5	
HOUST5F	Privately Owned Housing Starts: 5-Unit Structures or More (Thousands of Units)	5	
PERMIT	New Private Housing Units Authorized by Building Permits (Thousands of Units)	5	✓
HOUSTMW	Housing Starts in Midwest Census Region (Thousands of Units)	5	✓
HOUSTNE	Housing Starts in Northeast Census Region (Thousands of Units)	5	✓
HOUSTS	Housing Starts in South Census Region (Thousands of Units)	5	✓
HOUSTW	Housing Starts in West Census Region (Thousands of Units)	5	✓
USSTHPI	All-Transactions House Price Index for the United States (Index 1980 Q1=100)	5	✓
SPCS10RSA	S&P/Case-Shiller 10-City Composite Home Price Index (Index January 2000=100)	5	✓
SPCS20RSA	S&P/Case-Shiller 20-City Composite Home Price Index (Index January 2000=100)	5	✓
PERMITNE	New Private Housing Units Authorized by Building Permits: Northeast Census Region (Thousands, SAAR)	5	
PERMITMW	New Private Housing Units Authorized by Building Permits: Midwest Census Region (Thousands, SAAR)	5	
PERMITS	New Private Housing Units Authorized by Building Permits: South Census Region (Thousands, SAAR)	5	
PERMITW	New Private Housing Units Authorized by Building Permits: West Census Region (Thousands, SAAR)	5	

Group: Inventories, Orders, and Sales

Mnemonic	Description	Tcode	SW
CMRMTSPLx	Real Manufacturing and Trade Industries Sales (Millions of Chained 2012 Dollars)	5	
RSAFSx	Real Retail and Food Services Sales (Millions of Chained 2012 Dollars), deflated by Core PCE	5	✓
AMDMNOx	Real Manufacturers' New Orders: Durable Goods (Millions of 2012 Dollars), deflated by Core PCE	5	✓
ACOGNOx	Real Value of Manufacturers' New Orders for Consumer Goods Industries (Millions of 2012 Dollars), deflated by Core PCE	5	✓
AMDMUOx	Real Value of Manufacturers' Unfilled Orders for Durable Goods Industries (Millions of 2012 Dollars), deflated by Core PCE	5	✓
ANDENOx	Real Value of Manufacturers' New Orders for Capital Goods: Nondefense Capital Goods Industries (Millions of 2012 Dollars), deflated by Core PCE	5	✓
INVCQRMTSPL	Real Manufacturing and Trade Inventories (Millions of 2012 Dollars)	5	✓
BUSINVx	Total Business Inventories (Millions of Dollars)	5	
ISRATIOx	Total Business: Inventories to Sales Ratio	2	

Group: Prices

Mnemonic	Description	Tcode	SW
PCECTPI	Personal Consumption Expenditures: Chain-type Price Index (Index 2012=100)	6	
PCEPILFE	Personal Consumption Expenditures Excluding Food and Energy: Chain-type Price Index (Index 2012=100)	6	
GDPCTPI	Gross Domestic Product: Chain-type Price Index (Index 2012=100)	6	
GPDICTPI	Gross Private Domestic Investment: Chain-type Price Index (Index 2012=100)	6	✓
IPDBS	Business Sector: Implicit Price Deflator (Index 2012=100)	6	✓
DGDSRG3Q086SBEA	Personal Consumption Expenditures: Goods (Chain-type Price Index)	6	
DDURRG3Q086SBEA	Personal Consumption Expenditures: Durable Goods (Chain-type Price Index)	6	
DSERRG3Q086SBEA	Personal Consumption Expenditures: Services (Chain-type Price Index)	6	
DNDGRG3Q086SBEA	Personal Consumption Expenditures: Nondurable Goods (Chain-type Price Index)	6	
DHCERG3Q086SBEA	Personal Consumption Expenditures: Services: Household Consumption Expenditures (Chain-type Price Index)	6	

Mnemonic	Description	Tcode	SW
DMOTRG3Q086SBEA	Personal Consumption Expenditures: Durable Goods: Motor Vehicles and Parts (Chain-type Price Index)	6	✓
DFDHRG3Q086SBEA	Personal Consumption Expenditures: Durable Goods: Furnishings and Durable Household Equipment (Chain-type Price Index)	6	✓
DREQRG3Q086SBEA	Personal Consumption Expenditures: Durable Goods: Recreational Goods and Vehicles (Chain-type Price Index)	6	✓
DODGRG3Q086SBEA	Personal Consumption Expenditures: Durable Goods: Other Durable Goods (Chain-type Price Index)	6	✓
DFXARG3Q086SBEA	Personal Consumption Expenditures: Nondurable Goods: Food and Beverages Purchased for Off-Premises Consumption (Chain-type Price Index)	6	✓
DCLORG3Q086SBEA	Personal Consumption Expenditures: Nondurable Goods: Clothing and Footwear (Chain-type Price Index)	6	✓
DGOERG3Q086SBEA	Personal Consumption Expenditures: Nondurable Goods: Gasoline and Other Energy Goods (Chain-type Price Index)	6	✓
DONGRG3Q086SBEA	Personal Consumption Expenditures: Nondurable Goods: Other Nondurable Goods (Chain-type Price Index)	6	✓
DHUTRG3Q086SBEA	Personal Consumption Expenditures: Services: Housing and Utilities (Chain-type Price Index)	6	✓
DHLCRG3Q086SBEA	Personal Consumption Expenditures: Services: Health Care (Chain-type Price Index)	6	✓
DTRSRG3Q086SBEA	Personal Consumption Expenditures: Transportation Services (Chain-type Price Index)	6	✓
DRCARG3Q086SBEA	Personal Consumption Expenditures: Recreation Services (Chain-type Price Index)	6	✓
DFSARG3Q086SBEA	Personal Consumption Expenditures: Services: Food Services and Accommodations (Chain-type Price Index)	6	✓
DIFSRG3Q086SBEA	Personal Consumption Expenditures: Financial Services and Insurance (Chain-type Price Index)	6	✓
DOTSRG3Q086SBEA	Personal Consumption Expenditures: Other Services (Chain-type Price Index)	6	✓
CPIAUCSL	Consumer Price Index for All Urban Consumers: All Items (Index 1982-84=100)	6	
CPILFESL	Consumer Price Index for All Urban Consumers: All Items Less Food & Energy (Index 1982-84=100)	6	
WPSFD49207	Producer Price Index by Commodity for Finished Goods (Index 1982=100)	6	
PPIACO	Producer Price Index for All Commodities (Index 1982=100)	6	
WPSFD49502	Producer Price Index by Commodity for Finished Consumer Goods (Index 1982=100)	6	✓

Mnemonic	Description	Tcode	SW
WPSFD4111	Producer Price Index by Commodity for Finished Consumer Foods (Index 1982=100)	6	✓
PPIIDC	Producer Price Index by Commodity: Industrial Commodities (Index 1982=100)	6	✓
WPSID61	Producer Price Index by Commodity: Intermediate Materials, Supplies & Components (Index 1982=100)	6	✓
WPU0531	Producer Price Index by Commodity for Fuels and Related Products and Power: Natural Gas (Index 1982=100)	5	✓
WPU0561	Producer Price Index by Commodity for Fuels and Related Products and Power: Crude Petroleum, Domestic Production (Index 1982=100)	5	✓
OILPRICEx	Real Crude Oil Prices: West Texas Intermediate (WTI), Cushing, Oklahoma (2012 Dollars per Barrel), deflated by Core PCE	5	
WPSID62	Producer Price Index: Crude Materials for Further Processing (Index 1982=100)	6	
PPICMM	Producer Price Index: Commodities: Metals and Metal Products: Primary Nonferrous Metals (Index 1982=100)	6	
CPIAPPSL	Consumer Price Index for All Urban Consumers: Apparel (Index 1982-84=100)	6	
CPITRNSL	Consumer Price Index for All Urban Consumers: Transportation (Index 1982-84=100)	6	
CPIMEDSL	Consumer Price Index for All Urban Consumers: Medical Care (Index 1982-84=100)	6	
CUSR0000SAC	Consumer Price Index for All Urban Consumers: Commodities (Index 1982-84=100)	6	
CUSR0000SAD	Consumer Price Index for All Urban Consumers: Durables (Index 1982-84=100)	6	
CUSR0000SAS	Consumer Price Index for All Urban Consumers: Services (Index 1982-84=100)	6	
CPIULFSL	Consumer Price Index for All Urban Consumers: All Items Less Food (Index 1982-84=100)	6	
CUSR0000SA0L2	Consumer Price Index for All Urban Consumers: All Items Less Shelter (Index 1982-84=100)	6	
CUSR0000SA0L5	Consumer Price Index for All Urban Consumers: All Items Less Medical Care (Index 1982-84=100)	6	
CUSR0000SEHC	Consumer Price Index for All Urban Consumers: Owners' Equivalent Rent of Residences (Index Dec 1982=100)	6	

Group: Earnings and Productivity

Mnemonic	Description	Tcode	SW
AHETPIx	Real Average Hourly Earnings of Production and Nonsupervisory Employees: Total Private (2012 Dollars per Hour), deflated by Core PCE	5	
CES2000000008x	Real Average Hourly Earnings of Production and Nonsupervisory Employees: Construction (2012 Dollars per Hour), deflated by Core PCE	5	
CES3000000008x	Real Average Hourly Earnings of Production and Nonsupervisory Employees: Manufacturing (2012 Dollars per Hour), deflated by Core PCE	5	
COMPRMS	Manufacturing Sector: Real Compensation Per Hour (Index 2012=100)	5	✓
COMPRNFB	Nonfarm Business Sector: Real Compensation Per Hour (Index 2012=100)	5	✓
RCPHBS	Business Sector: Real Compensation Per Hour (Index 2012=100)	5	✓
OPHMFG	Manufacturing Sector: Real Output Per Hour of All Persons (Index 2012=100)	5	✓
OPHNFB	Nonfarm Business Sector: Real Output Per Hour of All Persons (Index 2012=100)	5	✓
OPHPBS	Business Sector: Real Output Per Hour of All Persons (Index 2012=100)	5	
ULCBS	Business Sector: Unit Labor Cost (Index 2012=100)	5	
ULCMFG	Manufacturing Sector: Unit Labor Cost (Index 2012=100)	5	✓
ULCNFB	Nonfarm Business Sector: Unit Labor Cost (Index 2012=100)	5	✓
UNLPNBS	Nonfarm Business Sector: Unit Nonlabor Payments (Index 2012=100)	5	✓
CES0600000008	Average Hourly Earnings of Production and Nonsupervisory Employees: Goods-Producing (Dollars per Hour)	6	

Group: Interest Rates

Mnemonic	Description	Tcode	SW
FEDFUNDS	Effective Federal Funds Rate (Percent)	2	✓
TB3MS	3-Month Treasury Bill: Secondary Market Rate (Percent)	2	✓
TB6MS	6-Month Treasury Bill: Secondary Market Rate (Percent)	2	
GS1	1-Year Treasury Constant Maturity Rate (Percent)	2	
GS10	10-Year Treasury Constant Maturity Rate (Percent)	2	
MORTGAGE30US	30-Year Conventional Mortgage Rate (Percent)	2	
AAA	Moody's Seasoned Aaa Corporate Bond Yield (Percent)	2	
BAA	Moody's Seasoned Baa Corporate Bond Yield (Percent)	2	

Mnemonic	Description	Tcode	SW
BAA10YM	Moody's Seasoned Baa Corporate Bond Yield Relative to Yield on 10-Year Treasury Constant Maturity (Percent)	1	✓
MORTG10YRx	30-Year Conventional Mortgage Rate Relative to 10-Year Treasury Constant Maturity (Percent)	1	✓
TB6M3Mx	6-Month Treasury Bill Minus 3-Month Treasury Bill, Secondary Market (Percent)	1	✓
GS1TB3Mx	1-Year Treasury Constant Maturity Minus 3-Month Treasury Bill, Secondary Market (Percent)	1	✓
GS10TB3Mx	10-Year Treasury Constant Maturity Minus 3-Month Treasury Bill, Secondary Market (Percent)	1	✓
CPF3MTB3Mx	3-Month Commercial Paper Minus 3-Month Treasury Bill, Secondary Market (Percent)	1	✓
GS5	5-Year Treasury Constant Maturity Rate (Percent)	2	
TB3SMFFM	3-Month Treasury Constant Maturity Minus Federal Funds Rate (Percent)	1	
T5YFFM	5-Year Treasury Constant Maturity Minus Federal Funds Rate (Percent)	1	
AAAFFM	Moody's Seasoned Aaa Corporate Bond Minus Federal Funds Rate (Percent)	1	
CP3M	3-Month AA Financial Commercial Paper Rate (Percent)	2	
COMPAPFF	3-Month Commercial Paper Minus Federal Funds Rate (Percent)	1	

Group: Money and Credit

Mnemonic	Description	Tcode	SW
BOGMBASEREALx	Real St. Louis Adjusted Monetary Base (Billions of 1982-84 Dollars), deflated by CPI	5	
M1REAL	Real M1 Money Stock (Billions of 1982-84 Dollars), deflated by CPI	5	
M2REAL	Real M2 Money Stock (Billions of 1982-84 Dollars), deflated by CPI	5	
BUSLOANSx	Real Commercial and Industrial Loans, All Commercial Banks (Billions of 2012 U.S. Dollars), deflated by Core PCE	5	✓
CONSUMERx	Real Consumer Loans at All Commercial Banks (Billions of 2012 U.S. Dollars), deflated by Core PCE	5	✓
NONREVSLx	Total Real Nonrevolving Credit Owned and Securitized, Outstanding (Billions of 2012 Dollars), deflated by Core PCE	5	✓

Mnemonic	Description	Tcode	SW
REALLNx	Real Real Estate Loans, All Commercial Banks (Billions of 2012 U.S. Dollars), deflated by Core PCE	5	✓
REVOLSLx	Total Real Revolving Credit Owned and Securitized, Outstanding (Billions of 2012 Dollars), deflated by Core PCE	5	✓
TOTALSLx	Total Consumer Credit Outstanding (Billions of 2012 Dollars), deflated by Core PCE	5	
DRIWCIL	Net Percentage of Domestic Banks Reporting Increased Willingness to Make Consumer Installment Loans (FRB Senior Loan Officer Opinion Survey)	1	✓
TOTRESNS	Total Reserves of Depository Institutions (Billions of Dollars)	6	
NONBORRES	Reserves of Depository Institutions, Nonborrowed (Millions of Dollars)	7	
DTCOLNVHFNM	Consumer Motor Vehicle Loans Outstanding Owned by Finance Companies (Millions of Dollars)	6	
DTCTHFNM	Total Consumer Loans and Leases Outstanding Owned and Securitized by Finance Companies (Millions of Dollars)	6	
INVEST	Securities in Bank Credit at All Commercial Banks (Billions of Dollars)	6	

Group: Household Balance Sheets

Mnemonic	Description	Tcode	SW
TABSHNOx	Real Total Assets of Households and Nonprofit Organizations (Billions of 2012 Dollars), deflated by Core PCE	5	
TLBSHNOx	Real Total Liabilities of Households and Nonprofit Organizations (Billions of 2012 Dollars), deflated by Core PCE	5	✓
LIABPIx	Liabilities of Households and Nonprofit Organizations Relative to Personal Disposable Income (Percent)	5	
TNWBSHNOx	Real Net Worth of Households and Nonprofit Organizations (Billions of 2012 Dollars), deflated by Core PCE	5	✓
NWPIx	Net Worth of Households and Nonprofit Organizations Relative to Disposable Personal Income (Percent)	1	
TARESAx	Real Assets of Households and Nonprofit Organizations Excluding Real Estate Assets (Billions of 2012 Dollars), deflated by Core PCE	5	✓

Mnemonic	Description	Tcode	SW
HNOREMQ027Sx	Real Real Estate Assets of Households and Nonprofit Organizations (Billions of 2012 Dollars), deflated by Core PCE	5	✓
TFAABSHNOx	Real Total Financial Assets of Households and Nonprofit Organizations (Billions of 2012 Dollars), deflated by Core PCE	5	✓
CONSPIx	Nonrevolving Consumer Credit to Personal Income (Percent)	2	

Group: Exchange Rates

Mnemonic	Description	Tcode	SW
TWEXAFEGSMTHx	Nominal Trade Weighted U.S. Dollar Index: Advanced Foreign Economies, Goods and Services (Index 2006=100)	5	✓
EXUSEU	U.S. / Euro Foreign Exchange Rate (U.S. Dollars to One Euro)	5	✓
EXSZUSx	Switzerland / U.S. Foreign Exchange Rate (Swiss Francs to One U.S. Dollar)	5	✓
EXJPUSx	Japan / U.S. Foreign Exchange Rate (Japanese Yen to One U.S. Dollar)	5	✓
EXUSUKx	U.S. / U.K. Foreign Exchange Rate (U.S. Dollars to One British Pound)	5	✓
EXCAUSx	Canada / U.S. Foreign Exchange Rate (Canadian Dollars to One U.S. Dollar)	5	✓

Group: Other

Mnemonic	Description	Tcode	SW
UMCSENTx	University of Michigan: Consumer Sentiment (Index 1st Quarter 1966=100)	1	✓
USEPUINDXM	Economic Policy Uncertainty Index for United States	2	✓

Group: Stock Markets

Mnemonic	Description	Tcode	SW
VIXCLSx	CBOE S&P 100 Volatility Index: VXO	1	✓
NIKKEI225	Nikkei Stock Average (Index)	5	
NASDAQCOM	NASDAQ Composite Index (Index Feb 5, 1971=100)	5	
S&P 500	S&P's Common Stock Price Index: Composite	5	✓
S&P div yield	S&P's Composite Common Stock: Dividend Yield (Percent)	2	
S&P PE ratio	S&P's Composite Common Stock: Price-Earnings Ratio	5	

Group: Non-Household Balance Sheets

Mnemonic	Description	Tcode	SW
GFDEGDQ188S	Federal Debt: Total Public Debt as Percent of GDP (Percent)	2	
GFDEBTNx	Real Federal Debt: Total Public Debt (Millions of 2012 Dollars), deflated by PCE	2	
TLBSNNCBx	Real Nonfinancial Corporate Business Sector Liabilities (Billions of 2012 Dollars), deflated by Implicit Price Deflator for Business Sector (IPDBS)	5	
TLBSNNCBBDIx	Nonfinancial Corporate Business Sector Liabilities to Disposable Business Income (Percent)	1	
TTAABSNNCBx	Real Nonfinancial Corporate Business Sector Assets (Billions of 2012 Dollars), deflated by Implicit Price Deflator for Business Sector (IPDBS)	5	
TNWMVBSNNCBx	Real Nonfinancial Corporate Business Sector Net Worth (Billions of 2012 Dollars), deflated by Implicit Price Deflator for Business Sector (IPDBS)	5	
TNWMVBSNNCBBDIx	Nonfinancial Corporate Business Sector Net Worth to Disposable Business Income (Percent)	2	
TLBSNNBx	Real Nonfinancial Noncorporate Business Sector Liabilities (Billions of 2012 Dollars), deflated by Implicit Price Deflator for Business Sector (IPDBS)	5	
TLBSNNBBDIx	Nonfinancial Noncorporate Business Sector Liabilities to Disposable Business Income (Percent)	1	
TABSNNBx	Real Nonfinancial Noncorporate Business Sector Assets (Billions of 2012 Dollars), deflated by Implicit Price Deflator for Business Sector (IPDBS)	5	
TNWBSNNBx	Real Nonfinancial Noncorporate Business Sector Net Worth (Billions of 2012 Dollars), deflated by Implicit Price Deflator for Business Sector (IPDBS)	5	

Mnemonic	Description	Tcode	SW
TNWBSNNBBDIx	Nonfinancial Noncorporate Business Sector Net Worth to Disposable Business Income (Percent)	2	
CNCFx	Real Disposable Business Income (Billions of 2012 Dollars); corporate cash flow with IVA minus taxes on corporate income, deflated by Implicit Price Deflator for Business Sector (IPDBS)	5	

B Austrian Data-Source Mapping

Table 16 reproduces the current Austria rows of `docs/data_sources.csv` (Section 3) – regenerated directly from that file every time this report is rebuilt, so it cannot drift out of sync with what `scripts/build_country_panel.R` actually resolved. 37 of 38 concepts were resolved as of the most recent run; an empty Provider/Key cell means the concept was attempted but no source returned data for Austria, with the reason given in the Comment column.

Table 16: Austria source mapping, regenerated from `docs/data_sources.csv`.

FRED-QD Group	Concept	Provider	Key	Comment
Output and Income	`real_gdp`	Eurostat	`namq_10_gdp:B1GQ (extended pre-1995 with level-spliced OECD_QNA:S1.B1GQ)`	
Output and Income	`real_household_consumption`	Eurostat	`namq_10_gdp:P31_S14 (extended pre-1995 with level-spliced OECD_QNA:S1M.P3)`	Source-dependent: for EU members (Eurostat NA_ITEM=P31_S14) this is household-only consumption, a close match to FRED-QD's household-only PCE; where OECD QNA is used instead (sector S1M), it is total-economy final consumption expenditure INCLUDING NPISHs, broader than FRED-QD's definition.

Table 16 continued

FRED-QD Group	Concept	Provider	Key	Comment
Output and Income	`real_govt_consumption`	Eurostat	`namq_10_gdp:P3_S13 (extended pre-1995 with level-spliced OECD_QNA:S13.P3)`	SNA/ESA transaction P3, sector S13 (general government) is government consumption expenditure only, whether sourced from OECD or Eurostat; FRED-QD's GCEC1 also includes government gross investment.
Output and Income	`real_gfcf_total`	Eurostat	`namq_10_gdp:P51G (extended pre-1995 with level-spliced OECD_QNA:.P51G)`	SNA/ESA transaction P51G (gross fixed capital formation) is for ALL sectors (incl. government), whether sourced from OECD or Eurostat; FRED-QD's FPIx is private-sector fixed investment only.
Output and Income	`real_exports`	Eurostat	`namq_10_gdp:P6 (extended pre-1995 with level-spliced OECD_QNA:S1.P6)`	
Output and Income	`real_imports`	Eurostat	`namq_10_gdp:P7 (extended pre-1995 with level-spliced OECD_QNA:S1.P7)`	
Output and Income	`real_household_disposable_income`	*unresolved*		Not resolved for AUT. OECD's quarterly household disposable income (DF_QNA_INC_SAV) is published for only 11 countries (AUS, BRA, CAN, CHL, EST, GRC, HUN, LTU, LUX, LVA, ZAF); absent for most others, confirmed absent for DEU/AUT/USA/FRA/GBR.
Industrial Production	`industrial_production`	FRED mirror	`AUTPROINDQISMEI`	

Table 16 continued

FRED-QD Group	**Concept**	**Provider**	**Key**	**Comment**
Industrial Production	`industrial_confidence`	EC Business/Consumer Survey	`AT.INDU`	EU member states only: European Commission Business and Consumer Survey, Industrial Confidence Indicator ("AT.INDU") – see R/ec_survey.R. No FRED-mirror fallback exists for this concept.
Employment and Unemployment	`unemployment_rate`	FRED mirror	`LRHUTTTTATQ156S`	
Employment and Unemployment	`employment_rate`	FRED mirror	`LREM64TTATQ156S`	Employment-to-population ratio, ages 15-64 (OECD MEI); included as a standard cross-country labour-market indicator even though FRED-QD has no direct equivalent (see us_note).
Employment and Unemployment	`employment_expectations`	EC Business/Consumer Survey	`AT.EEI`	EU member states only: European Commission Business and Consumer Survey, Employment Expectations Indicator ("AT.EEI") – see R/ec_survey.R. No FRED-mirror fallback exists for this concept.
Housing	`house_price_real`	FRED mirror	`QATR628BIS`	
Housing	`construction_confidence`	EC Business/Consumer Survey	`AT.BUIL`	EU member states only: European Commission Business and Consumer Survey, Construction Confidence Indicator ("AT.BUIL") – see R/ec_survey.R. No FRED-mirror fallback exists for this concept.
Inventories, Orders, and Sales	`retail_sales_volume`	FRED mirror	`AUTSARTQISMEI`	OECD MEI retail sales volume is not published for the USA itself via this mirror – a genuine coverage gap for that one country, not a wrong code.

Table 16 continued

FRED-QD Group	**Concept**	**Provider**	**Key**	**Comment**
Inventories, Orders, and Sales	`retail_confidence`	EC Business/Consumer Survey	`AT.RETA`	EU member states only: European Commission Business and Consumer Survey, Retail Trade Confidence Indicator ("AT.RETA") – see R/ec_survey.R. No FRED-mirror fallback exists for this concept.
Prices	`cpi_index`	Eurostat HICP	`prc_hicp_midx:M.I05.CP00.AT`	EU member states: sourced from Eurostat's Harmonised Index of Consumer Prices (HICP, all-items, prc_hicp_midx), NOT the frozen OECD-MEI-via-FRED mirror used for non-EU countries – see R/eurostat.R. Confirmed live 2026-08-30 to extend to 2025-Q4 for Austria, versus the FRED mirror's confirmed freeze at 2023-Q4 – a roughly 2-year improvement. HICP's basket/methodology differs somewhat from the US CPI-U basket underlying FRED-QD's CPIAUCSL, but is the standard EU consumer-price measure. Falls back to the FRED mirror if the Eurostat series is unavailable for a given run.
Prices	`core_cpi_index`	Eurostat HICP	`prc_hicp_midx:M.I05.TOT_X_NRG_FOOD.AT`	Eurostat HICP excluding energy, food, alcohol and tobacco (COICOP=TOT_X_NRG_FOOD) – the standard ECB/Eurostat "core inflation" measure. EU member states only; no FRED-mirror fallback exists for this concept.

Table 16 continued

FRED-QD Group	Concept	Provider	Key	Comment
Prices	`food_price_index`	Eurostat HICP	`prc_hicp_midx:M.I05.CP01.AT`	Eurostat HICP, food and non-alcoholic beverages (COICOP=CP01). EU member states only; no FRED-mirror fallback exists for this concept.
Prices	`energy_price_index`	Eurostat HICP	`prc_hicp_midx:M.I05.NRG.AT`	Eurostat HICP, energy (COICOP=NRG). EU member states only; no FRED-mirror fallback exists for this concept.
Prices	`services_price_index`	Eurostat HICP	`prc_hicp_midx:M.I05.SERV.AT`	Eurostat HICP, services (overall index excluding goods) (COICOP=SERV). EU member states only; no FRED-mirror fallback exists for this concept.
Earnings and Productivity	`unit_labor_cost`	Eurostat ULC	`namq_10_lp_ulc:Q.I10.SCA.NULC_HW.AT`	Where Eurostat publishes an index-level series for it (confirmed for Austria: namq_10_lp_ulc, NA_ITEM=NULC_HW, UNIT=I10, hours-based like FRED-QD's ULCNFB), this replaces the default OECD-mirror proxy (OECD MEI unit labour cost, employment-based, % change, confirmed live for AT/DE/FR/GB/US) – see R/eurostat.R. Not every EU country publishes this index-level series (confirmed absent for Germany, which keeps the OECD-mirror value).
Interest Rates	`long_term_rate`	FRED mirror	`IRLTLT01ATQ156N`	
Interest Rates	`short_term_rate`	FRED mirror	`IR3TIB01ATQ156N`	

Table 16 continued

FRED-QD Group	Concept	Provider	Key	Comment
Interest Rates	`mortgage_rate`	ECB MIR	`A2C.R.A.2250.EUR.N`	ECB MFI Interest Rate Statistics (MIR): new-business loans to households for house purchase, all initial rate fixation periods combined – genuinely country-specific (unlike euro_area_household_net_worth_growth above), available for euro-area members only.
Money and Credit	`credit_to_private_nonfin_sector`	BIS WS_TC	`P`	BIS reports this as a stock, % of GDP (private non-financial sector = households + nonfinancial corporations combined).
Money and Credit	`household_mortgage_loans`	ECB BSI	`A22T.A.1.U6.2250.Z01.E`	ECB MFI Balance Sheet Items (BSI): outstanding amounts (stocks, millions of EUR) of loans to households for house purchase, domestic counterpart – genuinely country-specific (unlike euro_area_household_net_worth_growth), available for euro-area members only. Same purpose category as mortgage_rate, but a different ECB dataflow with an entirely different dimension structure – see R/ecb.R.
Household Balance Sheets	`euro_area_household_net_worth_growth`	ECB QSA_PUB	`I8`	ECB QSA_PUB publishes household net worth only for the euro-area AGGREGATE (REF_AREA=I8) – every euro-area country gets this same figure; it is not country-specific.
Household Balance Sheets	`household_credit_to_gdp`	BIS WS_TC	`H`	BIS credit to households & NPISHs, % of GDP – country-specific (unlike the ECB net-worth aggregate above).
Non-Household Balance Sheets	`corporate_credit_to_gdp`	BIS WS_TC	`N`	BIS credit to nonfinancial corporations, % of GDP – country-specific.

Table 16 continued

FRED-QD Group	Concept	Provider	Key	Comment
Non-Household Balance Sheets	`government_debt_to_gdp`	BIS WS_TC	`G`	BIS credit to general government (all levels: federal/state/local combined), % of GDP – BIS's own credit-statistics methodology treats this as a close proxy for gross government debt; genuinely country-specific and, unlike the euro-area-only mortgage_rate/consumer_confidence overrides, available for non-EU countries too (confirmed live for AT/DE/US).
Exchange Rates	`fx_rate_to_usd`	FRED mirror	`CCUSMA02ATQ618N`	OECD MEI bilateral exchange rate, national currency per USD.
Exchange Rates	`real_effective_exchange_rate`	FRED mirror	`CCRETT01ATQ661N`	OECD real (price-adjusted) effective exchange rate index – see us_note for how this differs from FRED-QD's nominal series.
Other	`consumer_confidence`	EC Business/Consumer Survey	`AT.CONS`	EU member states: sourced from the European Commission's own Business and Consumer Survey (a live, monthly, seasonally adjusted balance statistic, e.g. "AT.CONS"), NOT the frozen OECD-MEI-via-FRED mirror used for non-EU countries – see R/ec_survey.R. Falls back to the FRED mirror if the EC archive is unavailable for a given run.
Other	`economic_sentiment_indicator`	EC Business/Consumer Survey	`AT.ESI`	EU member states only: European Commission Business and Consumer Survey, Economic Sentiment Indicator ("AT.ESI") – see R/ec_survey.R. No FRED-mirror fallback exists for this concept.

Table 16 continued

FRED-QD Group	Concept	Provider	Key	Comment
Other	`services_confidence`	EC Business/Consumer Survey	`AT.SERV`	EU member states only: European Commission Business and Consumer Survey, Services Confidence Indicator (”AT.SERV”) – see R/ec_survey.R. No FRED-mirror fallback exists for this concept.
Other	`geopolitical_risk`	GPR Index	`GPR`	Caldara and Iacoviello’s (2022) Geopolitical Risk index, from matteoiacoviello.com’s own published data file – see R/gpr.R. Genuinely country-specific for the 44 countries the source constructs one for (confirmed: Germany, the United States); the global index is used for every other country (confirmed: Austria), not a country-specific gap in this project’s own sourcing.
Stock Markets	`share_price_index`	Yahoo Finance	`^ATX`	Austria: sourced from the ATX (Austrian Traded Index) via Yahoo Finance (ticker ”ˆATX”), Austria’s own actual benchmark index, NOT the generic OECD MEI ’all shares’ proxy used for other countries – see R/yahoo_finance.R. Falls back to the FRED mirror if the Yahoo Finance fetch is unavailable for a given run.